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National freeways and electronic tolling

The 國道 network's numbered routes across three self-described construction generations, and the 2013 switch to all-electronic distance-based tolling that laid off 947 staff in one day.

NATIONAL FREEWAYS AND ELECTRONIC TOLLING

NATIONAL HIGHWAY 1 (中山高速公路), OPENED 1978, built in 7 years [1]	NATIONAL HIGHWAY 5 (蔣渭水高速公路), OPERATIONAL SECTION 55 km, Nangang [/rail/thsr/stations/nangang/] (Taipei [/rail/thsr/stations/taipei/]) to Su'ao (Yilan) [1]	XUESHAN TUNNEL COMPLETED 2006-06-16, approx. 12.9 km, stated cost NT$90.6 billion [1]	FIRST TOLLBOOTH 1974-07-30, Taishan [3]
PER-TRIP ELECTRONIC TOLLING INTRODUCED ALONGSIDE MANUAL LANES 2006-02-10 [3]	FULL SWITCH TO ALL-ELECTRONIC, DISTANCE-BASED TOLLING 2013-12-30, manned tollbooths closed [3]	TOLL-COLLECTION STAFF LET GO IN THE SWITCH 947, per the Freeway Bureau [/road/network/national-freeways/]'s own account [4]	FETC'S NEWEST ETC CONTRACT 10 years, 2024-12-22 to 2034-12-31; per-km rate cut from NT$0.070143 to NT$0.062; BOT converted to ROT [6]
NATIONAL HIGHWAY 7, STATUS not yet operational — still in design/tendering as of the Freeway Bureau [/road/network/national-freeways/]'s own page (stated last updated 2025-11-20) [5]			

Taiwan's 國道 freeway network is built and maintained by 交通部高速公路局 (the Freeway Bureau), successor to the former National Freeway Construction Bureau. [1] The Bureau's own history page frames the network in three self-described construction generations: a first generation

(National Highway 1, built to a basic construction standard), a second generation (Highways 2, 3, 4, 5, 8 and 10, adding landscape and aesthetic design work), and a third generation (Highway 6 alone, which the Bureau says was built to prioritise energy efficiency and sustainable development, with ecological corridors and curved routing through sensitive terrain rather than straight-line tunnels).[1]

The route list, and a length that does not agree with itself

National Highway 1, the 中山高速公路 (Chung-Shan Freeway), opened in 1978 after seven years of construction — the Bureau's own history page states this directly: 「長達372.5公里的中山高速公路，在7年之內便興建完工」.[1] That same sentence gives the route's length as 372.5 km. The Bureau's own current interchange-and-service-area mileage table for the same highway, however, shows the route terminating at kilometre marker 373, in a table row reading 「高雄端 | 373 | 高雄市中山四路、高雄航空站」.[2] Both figures come from the same agency's own published material and this page does not pick between them — they may be measuring the original alignment against a current mileage-marker terminus, but neither page says so, and the 0.5 km gap is published here as an open conflict rather than resolved.

National Highway 5, the 蔣渭水高速公路 (Chiang Wei-shui Freeway), has an operational section of 55 km from Nangang in Taipei `[/rail/thsr/stations/taipei/]` to Su'ao in Yilan.[1] It contains the Xueshan Tunnel, roughly 12.9 km between Pinglin and Toucheng, completed 16 June 2006 at a stated cost of NT$90.6 billion after a 15-year project.[1] National Highway 6, the 水沙連高速公路, is the Bureau's own example of its third-generation design philosophy: roughly 38 km branching from Highway 3 at Wufeng through Caotun and Guoxing to Puli.[1]

National Highway 7, a planned Kaohsiung route from Renwu to Xiaogang, is not yet built. The Bureau's own project page, stated last updated 20 November 2025, puts it at roughly 23 km and NT$150.17 billion, still in design-and-tendering across six civil-works packages rather than under construction.[5] Its absence from the operational mileage tables covering the other numbered highways is consistent with that stated status, not an omission.

Electronic tolling: from the first tollbooth to no tollbooths at all

Taiwan's first freeway tollbooth opened at Taishan on 30 July 1974, per the Bureau's own account of a network that had until then collected no tolls.[3] Electronic tolling was introduced gradually rather than all at once: a per-trip ETC lane system began running alongside manual booths on 10 February 2006, under a BOT contract structure adopted in 2003.[3] The full switch to distance-based, all-electronic tolling — the end of every staffed tollbooth on the network — came on 30 December 2013.[3]

That switch had a direct workforce consequence the Bureau states plainly on its own site: 947 toll-collection staff were let go in the transition.[4] The ETC contract required the operator, 遠通電收 (Far Eastern Electronic Toll Collection, FETC), to fully absorb affected staff into new roles by a Bureau-set deadline of 30 June 2014, and the Bureau's own page states it imposed a NT$4.5 million contractual penalty for late completion, with FETC ultimately prevailing in the resulting litigation over whether the obligation had been fulfilled.[4]

FETC's ETC concession has since been renewed. Per CNA reporting, the new contract runs from 22 December 2024 to 31 December 2034, cuts the delegated per-vehicle-kilometre fee from NT$0.070143 to NT$0.062, and is projected to cut the government's own annual payment to FETC from roughly NT$22.6 billion to roughly NT$2 billion at current traffic volumes — a contract that also converts the underlying model from BOT to ROT, meaning the tolling assets themselves transfer to government ownership.[6]

What remains TBC

A single Bureau-published figure for the network's total combined length across every numbered route was not located during this research. The exact current total number of freeway service areas is likewise TBC — the Bureau's own service-area page did not carry a single explicit "total" sentence in the version fetched for this research. Reports in secondary sources of legal disputes over the original 2003 ETC contract award and of later usage-rate-target penalty disputes between the Bureau and FETC were found during this research but not independently verified against a primary 審計部 or 監察院 document, and are not published here as established fact for that reason.

Specifications

National Highway 1 length (Freeway Bureau history page)	**372.5**	km[1]
National Highway 1 length (Freeway Bureau mileage table)	**373**	km[2]
National Highway 6 (水沙連高速公路) length	**38**	km (approximate, per the Bureau's own history page) [1]

References

5 of 6 sources on this page are primary — published by the operator, the builder or government.

1 Freeway network history [https://www.freeway.gov.tw/Publish.aspx?cnid=3005]
PRIMARY 交通部高速公路局（Freeway Bureau, MOTC）accessed 2026-08-29
The Bureau's own history page. States National Highway 1 (中山高速公路) as 372.5 km, opened 1978 after 7 years' construction: 「長達372.5公里的中山高速公路，在7年之內便興建完工」,「於民國67年通車」. Frames the network in three generations: first (Highway 1, basic construction standard), second (Highways 2, 3, 4, 5, 8, 10, adding landscape design), third (Highway 6, prioritising energy

efficiency and sustainable development). States Highway 5's operational section as 55 km, Nangang to Su'ao, and the Xueshan Tunnel (12.9 km, Pinglin-Toucheng) as completed 16 June 2006 at a stated cost of NT$90.6 billion. States Highway 6 (水沙連高速公路) as approximately 38 km, Wufeng to Puli.

2 National Highway 1 interchange and service-area mileage table [https://www.freeway.gov.tw/Publish.aspx?cnid=1906&p=4617]
PRIMARY 交通部高速公路局（Freeway Bureau, MOTC）accessed 2026-08-29
The Bureau's current operational mileage table for National Highway 1, terminating at km 373: 「高雄端 | 373 | 高雄市中山四路、高雄航空站」— a different figure from the same Bureau's own history page (372.5 km).

3 ETC introduction timeline [https://www.freeway.gov.tw/Publish.aspx?cnid=1482&p=509]
PRIMARY 交通部高速公路局（Freeway Bureau, MOTC）accessed 2026-08-29
The Bureau's own ETC timeline: first tollbooth 30 July 1974 at Taishan (「63年7月30日泰山收費站...為第一個開始收費的收費站」); per-trip ETC lanes introduced 10 February 2006 running parallel to manual booths (「95年2月10日啟用計次電子收費(ETC)車道，人工及電子收費併行」); full switch to distance-based all-electronic tolling 30 December 2013, when manned tollbooths closed (「102年12月30日計次人工及電子收費轉換為全電子計程收費」); BOT model for ETC adopted 2003.

4 Toll-collection staff transition [https://www.freeway.gov.tw/Publish.aspx?cnid=133]
PRIMARY 交通部高速公路局（Freeway Bureau, MOTC）accessed 2026-08-29
The Bureau's own account of the 2013 all-electronic switch: 「高公局收費人員共計947人全數精簡」— 947 toll-collection staff let go. States the FETC contract required the operator to fully absorb affected staff by 30 June 2014 (「須於103年6月30日前應完成所有收費員全數吸收轉置」) and that a contractual penalty of NT$4.5 million was imposed for late completion, with FETC ultimately prevailing in the related litigation.

5 National Highway 7 project page [https://www.freeway.gov.tw/Publish.aspx?cnid=95&p=8924]
PRIMARY 交通部高速公路局（Freeway Bureau, MOTC）accessed 2026-08-29
The Bureau's own project page, stated last updated 20 November 2025 (114年11月20日). States National Highway 7 as approximately 23 km, Kaohsiung Renwu to Xiaogang, budgeted at approximately NT$150.17 billion, and as of the page's update still in design/tendering across 6 civil-works packages (「目前分6土建標辦理設計作業，陸續辦理招標作業中」) — not under construction.

6 FETC's ETC contract renewed for 10 years, with a lower per-km rate and new eTag incentive [https://www.cna.com.tw/news/ahel/202512170152.aspx]
ETC由遠通續約10年 新約首年新申辦eTag送儲值金
SECONDARY 中央社（Central News Agency, CNA）accessed 2026-08-29
Reports FETC's new contract runs 22 December 2024 to 31 December 2034; the per-vehicle-kilometre delegated fee falls from NT$0.070143 to NT$0.062; the government's projected annual payment falls from roughly NT$22.6 billion to roughly NT$2 billion at current traffic volumes; the contract model converts from BOT to ROT (assets transfer to government); toll-gate-to-control-centre data transmission time is to fall from 7 minutes to 2 minutes; new eTag sign-ups in the contract's first year that activate automatic bank payment within a month receive NT$100 in stored value.

Last updated: 2026-08-29

Station names, codes and sequence from TDX Transport Data eXchange, Ministry of Transportation and Communications, operators TRTC, NTMC, TYMC, and TMRT. Government open data. Retrieved 2026-08-24; source last updated 2016-11-23. The 12 Sanying station records absent from TDX are sourced directly to New Taipei Metro and government publications on their pages. Provenance.